

Final Capstone Project: Azure Databricks

End-to-End Retail Analytics Platform

Business Scenario

A retail company wants to build a complete analytics platform using **Azure Databricks**. The system should process raw files, clean data, create Delta tables, perform transformations, build Gold reports, visualize insights, and schedule the workflow as a production job.

Datasets Required

1. customers.csv

```
customer_id,customer_name,city,state,customer_type
C101,Rahul Sharma,Hyderabad,Telangana,Premium
C102,Priya Reddy,Bangalore,Karnataka,Regular
C103,Amit Kumar,Mumbai,Maharashtra,Regular
C104,Sneha Patel,Delhi,Delhi,Premium
C105,Farhan Ali,Chennai,Tamil Nadu,Regular
C106,Neha Singh,Pune,Maharashtra,Premium
C107,Arjun Verma,Hyderabad,Telangana,Regular
C108,Meera Nair,Kochi,Kerala,Premium
```

2. products.csv

```
product_id,product_name,category,unit_price
P101,Laptop,Electronics,65000
P102,Mobile,Electronics,25000
P103,Chair,Furniture,7000
P104,Table,Furniture,12000
P105,Shoes,Fashion,4500
P106,Watch,Fashion,8000
```

P107,TV,Electronics,45000

P108,Bag,Fashion,3000

3. orders.csv

```
order_id,customer_id,product_id,order_date,quantity,status
O1001,C101,P101,2026-06-01,1,Completed
O1002,C102,P102,2026-06-01,2,Completed
O1003,C103,P103,2026-06-02,3,Pending
O1004,C104,P104,2026-06-02,2,Completed
O1005,C105,P105,2026-06-03,4,Cancelled
O1006,C106,P106,2026-06-03,1,Completed
O1007,C107,P107,2026-06-04,1,Completed
O1008,C108,P108,2026-06-04,5,Completed
O1009,C101,P102,2026-06-05,1,Completed
O1010,C104,P101,2026-06-05,1,Completed
```

4. customer_preferences.json

```
[
  {
    "customer_id": "C101",
    "preferred_channel": "Online",
    "loyalty": {
      "tier": "Gold",
      "points": 1200
    }
  },
  {
    "customer_id": "C102",
    "preferred_channel": "Store",
    "loyalty": {
      "tier": "Silver",
      "points": 700
    }
  },
  {
    "customer_id": "C104",
    "preferred_channel": "Online",
    "loyalty": {
```

```
        "tier": "Platinum",
        "points": 2200
    },
    {
        "customer_id": "C108",
        "preferred_channel": "Mobile App",
        "loyalty": {
            "tier": "Gold",
            "points": 1500
        }
    }
]
```

Capstone Requirements

Part 1: Ingestion

1. Upload CSV and JSON files into Databricks.
 2. Read customers.csv.
 3. Read products.csv.
 4. Read orders.csv.
 5. Read customer_preferences.json.
 6. Display schema of all datasets.
 7. Save raw data as Bronze Delta tables.
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Part 2: Cleaning and Transformation

8. Handle null values if present.
9. Flatten customer_preferences.json.
10. Join customers with preferences.
11. Join orders with customers.
12. Join orders with products.
13. Create revenue = quantity * unit_price.
14. Create order_month from order_date.
15. Create customer_segment:
Premium -> High Value

Regular -> Standard Value

16. Save cleaned data as Silver Delta tables.

Part 3: Spark SQL

- 17. Create temporary views.
 - 18. Write SQL query for total revenue.
 - 19. Write SQL query for revenue by city.
 - 20. Write SQL query for revenue by category.
 - 21. Write SQL query for top customers.
 - 22. Write SQL query for completed orders only.
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Part 4: Window Functions

- 23. Rank products by revenue.
 - 24. Rank customers by revenue.
 - 25. Find top 3 customers.
 - 26. Find top product in each category.
 - 27. Create running revenue by order_date.
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Part 5: Delta Lake

- 28. Create Delta table using DataFrame write.
 - 29. Create Delta table using saveAsTable().
 - 30. Create Delta table using SQL.
 - 31. View Delta history.
 - 32. Read previous Delta version using Time Travel.
 - 33. Perform SCD Type 1 merge using updated customer data.
 - 34. Verify version increment after merge.
 - 35. Run OPTIMIZE.
 - 36. Run ZORDER BY(city).
 - 37. Run VACUUM.
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Part 6: Visualization

- 38. Bar chart: Revenue by City.
 - 39. Bar chart: Revenue by Category.
 - 40. Pie chart: Orders by Status.
 - 41. Horizontal bar chart: Top Products by Revenue.
 - 42. Line chart: Revenue Trend by Date.
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Part 7: Tables and Views

- 43. Create managed table.
 - 44. Create external table.
 - 45. Create temporary view.
 - 46. Create global temporary view.
 - 47. Compare scope and lifetime of each.
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